

BUFFALO, MN, USA

WMO: 722114

Lat: 45.159N

Lon: 93.843W

Elev: 295

StdP: 97.83

Time Zone: -6.00 (NAC)

Period: 04-19

WBAN: 54901

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-24.9	-22.5	-28.8	0.3	-24.4	-26.6	0.3	-22.1	9.3	-8.7	8.4	-7.3	1.9	280	0.518

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	10.3	31.3	22.6	29.9	21.7	28.5	20.7	24.5	29.4	23.3	28.2	22.2	26.7	3.7	170

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
22.8	18.1	28.0	21.6	16.9	27.0	20.5	15.8	25.5	75.4	29.5	70.8	28.2	66.6	26.6	29.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
8.7	7.7	7.0	DB	-29.1	34.1	2.6	1.6	-31.0	35.2	-32.5	36.2	-34.0	37.1	-35.9	38.3	
			WB	-29.3	26.2	2.6	1.3	-31.1	27.1	-32.7	27.9	-34.1	28.6	-36.0	29.6	

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	7.3	-9.5	-8.3	-0.1	7.3	14.6	20.0	22.6	20.9	16.9	8.9	0.8	-6.8	(6)
	DBStd	12.57	7.32	6.68	6.92	5.59	4.94	3.25	2.96	2.79	4.45	4.94	5.89	6.45	(7)
	HDD10.0	2452	604	513	321	114	14	0	0	0	3	81	279	522	(8)
	HDD18.3	4363	863	747	573	331	138	19	3	8	78	296	526	780	(9)
	CDD10.0	1484	0	0	7	33	155	301	389	337	211	46	4	0	(10)
	CDD18.3	353	0	0	1	1	21	70	134	87	37	3	0	0	(11)
	CDH23.3	2913	0	0	2	12	221	562	1171	644	276	27	0	0	(12)
	CDH26.7	792	0	0	0	1	67	143	364	149	64	3	0	0	(13)
(14) Wind	WSAvg	2.8	2.9	2.9	3.1	3.6	3.2	2.7	2.3	2.1	2.6	2.9	3.0	2.9	(14)
(15) Precipitation	PrecAvg	797	19	18	39	72	93	121	98	108	92	65	41	23	(15)
	PrecMax	1292	50	42	91	202	196	338	263	217	211	168	120	57	(16)
	PrecMin	545	2	3	8	16	21	48	25	17	15	15	2	6	(17)
	PrecStd	158	14	11	21	43	43	63	57	54	56	41	31	15	(18)
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	5.9	11.0	20.7	25.0	32.4	32.5	34.5	32.1	31.0	26.8	18.9	8.7	(19)
		MCWB	2.9	8.3	14.2	14.5	20.2	22.9	25.1	23.4	22.2	18.0	12.8	6.0	(20)
	2%	DB	3.4	5.3	16.0	21.5	28.2	30.0	31.7	29.9	28.5	22.9	14.8	4.9	(21)
		MCWB	1.5	2.8	11.0	12.5	18.5	21.0	23.6	22.4	20.8	15.9	10.3	2.7	(22)
	5%	DB	1.8	2.8	12.7	19.1	25.6	28.2	30.1	28.3	26.8	19.7	11.7	2.8	(23)
		MCWB	0.5	1.1	9.1	11.6	17.1	19.8	22.3	20.8	19.9	13.7	7.9	1.5	(24)
	10%	DB	0.2	0.9	9.1	16.7	22.9	26.7	28.7	26.9	24.5	16.9	9.1	1.3	(25)
		MCWB	-0.6	-0.5	6.3	10.3	15.6	19.0	21.5	20.2	18.7	12.0	6.1	0.4	(26)
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	3.1	8.3	16.3	15.8	23.4	24.4	26.7	24.9	23.6	20.3	13.8	7.6	(27)
		MCDB	5.4	11.5	20.1	21.6	28.6	30.4	31.6	29.7	28.4	23.2	17.2	8.6	(28)
	2%	WB	1.8	3.1	12.3	14.1	20.2	22.8	25.1	23.7	21.9	16.6	10.7	3.0	(29)
		MCDB	3.1	4.9	15.2	19.6	25.9	27.9	29.8	28.3	26.8	21.5	14.3	4.1	(30)
	5%	WB	0.6	1.1	9.0	12.5	18.3	21.5	23.7	22.5	20.7	14.6	8.1	1.6	(31)
		MCDB	1.6	2.6	12.4	17.8	23.0	26.3	28.7	26.6	25.1	18.7	11.4	2.7	(32)
	10%	WB	-0.5	-0.3	6.2	10.8	16.9	20.1	22.5	21.4	19.5	12.5	6.1	0.5	(33)
		MCDB	0.3	0.7	9.0	15.7	21.8	24.7	27.4	25.4	24.0	16.2	9.0	1.3	(34)
(35) Mean Daily Temperature Range	5% DB	MDBR	8.5	9.5	9.3	10.2	10.7	10.0	10.3	10.1	10.5	9.3	8.2	7.4	(35)
		MCDBR	9.3	10.5	12.7	14.6	14.7	12.2	11.4	11.4	12.4	13.7	12.0	8.3	(36)
	5% WB	MCWBR	7.8	8.4	8.3	8.3	6.8	5.6	5.3	5.5	6.1	7.6	8.0	7.1	(37)
		MCWBR	9.1	9.7	11.4	12.5	12.1	10.4	10.4	9.9	10.9	11.3	11.4	7.4	(38)
(40) Clear-Sky Solar Irradiance	taub		0.266	0.278	0.313	0.369	0.403	0.394	0.393	0.401	0.371	0.317	0.287	0.263	(40)
		taud	2.416	2.373	2.395	2.308	2.267	2.338	2.364	2.334	2.384	2.534	2.513	2.441	(41)
	Ebn at Noon		860	914	923	890	867	874	870	849	846	860	831	828	(42)
		Edh at Noon	71	90	102	122	131	123	118	118	103	77	65	63	(43)
(44) All-Sky Solar Radiation	RadAvg		1.73	2.81	3.81	4.63	5.35	5.95	6.34	5.28	4.10	2.60	1.66	1.22	(44)
	RadStd		0.20	0.29	0.35	0.44	0.43	0.45	0.23	0.40	0.29	0.35	0.15	0.14	(45)

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46)	Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(47)	Regional (63 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page